

CS23334 Fundamentals of Data Science Lab

Experiment 9

Understanding Matplotlib & Seaborn Library Use Cases in Data Science

Objective: To explore, analyze, and visualize multi-dimensional feature distributions and relationships in the Iris dataset using Matplotlib and Seaborn.

Methodology:

- 1 **Data Ingestion & Inspection:** Load the Iris dataset and inspect summary statistics, data types, and variety distributions.
- 2 **Categorical & Bivariate Visualization:** Use count plots and scatter plots to analyze class balances and dimension separations.
- 3 **Multivariate Pairplot:** Generate pairwise scatter and kernel density plots to capture multi-attribute clustering patterns.
- 4 **Faceted Histograms:** Construct conditional distribution grids across all sepal and petal features grouped by variety.

Code:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_csv('Iris_Dataset.csv')

print(data.info())
print(data.describe())
print(data.value_counts('variety'))

sns.countplot(x='variety', data=data)
plt.show()

dummies = pd.get_dummies(data.variety)
FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)
print(FinalDataset.head())

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data)
plt.show()

sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data)
plt.show()

sns.pairplot(data, hue='variety', height=3)
plt.show()

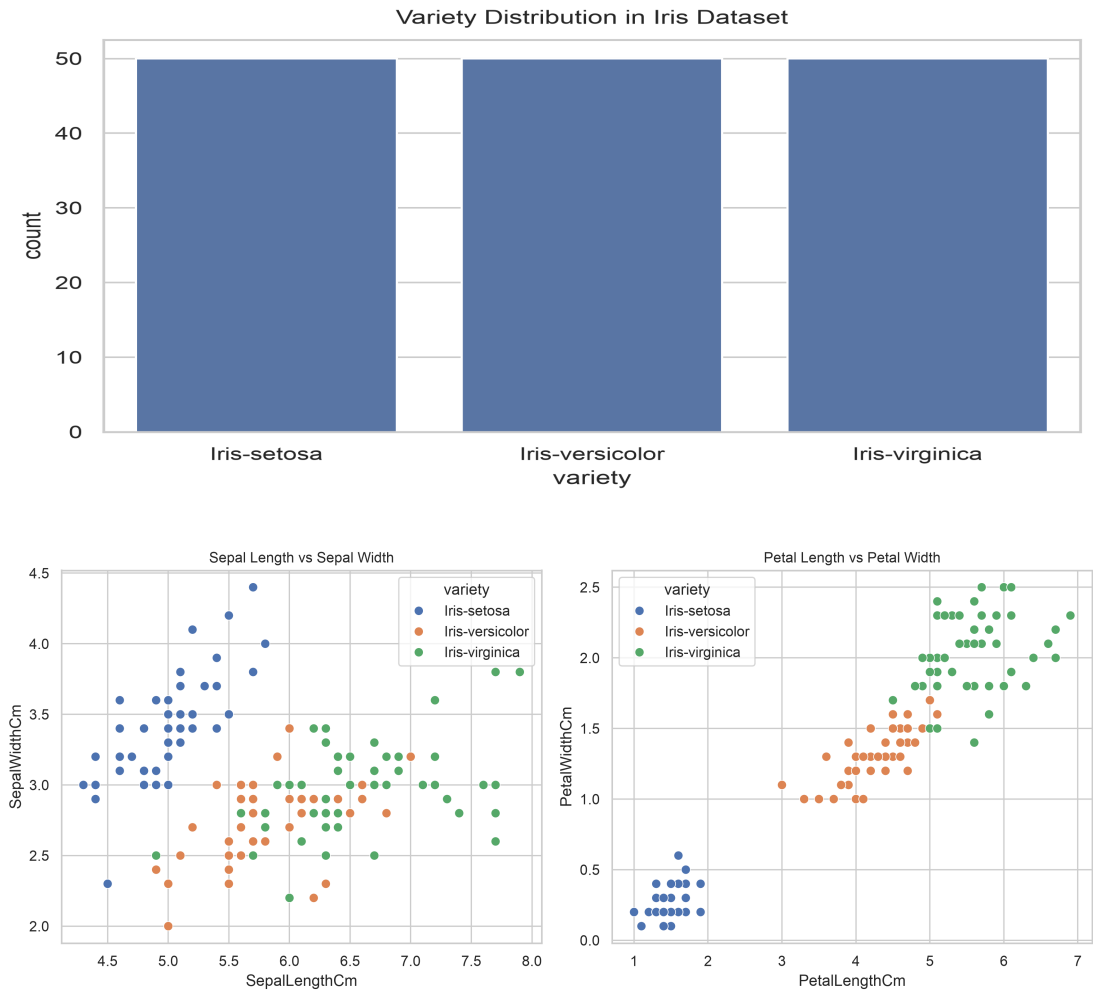
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalLengthCm').add_legend()
plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalWidthCm').add_legend()
plt.show()

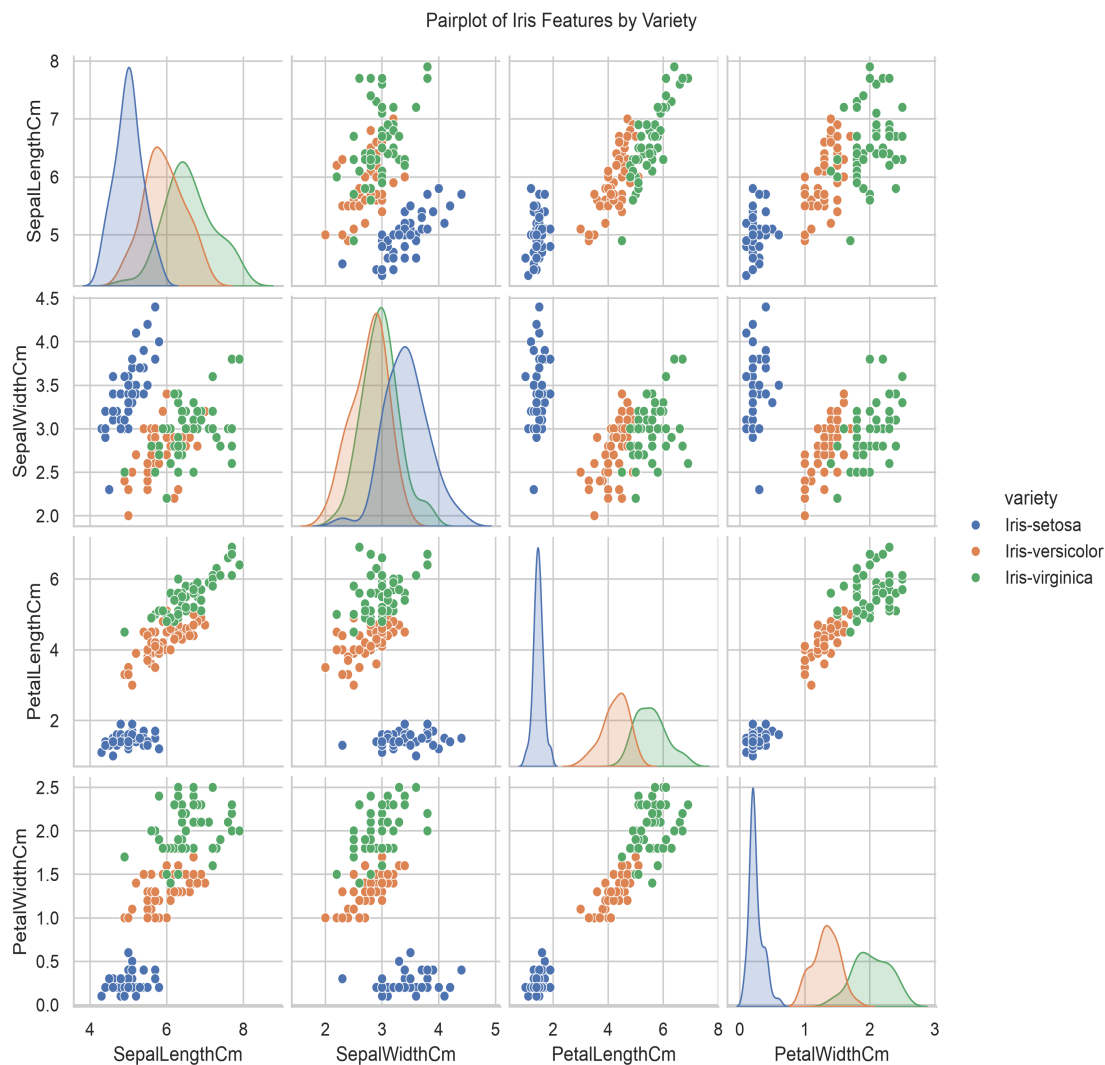
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalLengthCm').add_legend()
plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalWidthCm').add_legend()
plt.show()
```

OUTPUT: 1. Categorical Distribution & Bivariate Scatter Plots



OUTPUT: 2. Multivariate Pairplot Analysis



OUTPUT: 3. Feature Distributions (FacetGrid Histograms)

